

Recipes for Creating Digital Game Prototypes

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ABSTRACT

This lab introduces aspiring or current game designers with limited programming knowledge, to concepts that can be used to create digital game prototypes. Four basic concepts are presented as ingredients (vectors, time, collisions, and lines and rays), each with explanations and examples demonstrated in Unreal and Unity. Then examples are provided on how the ingredients can be combined into recipes to create common gameplay mechanics for prototyping. This is not an exhaustive list of everything someone needs to develop digital game prototypes, however, it serves as a good starting point for beginners who may want to start designing and prototyping games with limited knowledge of programming in game engines. This lab assumes a basic understanding of Unreal, Unity, or a comparable game engine, but programming experience is not required. Upon completion of this lab, participants should have several recipes they can use, build on, and modify for creating simple game play mechanics for prototyping. The example projects used in the lab will be available for download.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts.

KEYWORDS

prototyping, game design, game engine

ACM Reference Format:

Justin Johnson. 2024. Recipes for Creating Digital Game Prototypes. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH Labs '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641236.3665165>

1 INTRODUCTION

Prototyping is an important part of designing games. It allows designers to rapidly test ideas and game concepts before moving into time consuming production. Depending on the purpose, or what is being tested, prototypes can come in many forms: physical (analog) or digital, rapid low fidelity or polished vertical slice, and everything in between. Analog prototypes are common in early stages of game development and can be made with almost any material on hand, requiring a low skill barrier. A digital prototype, on the other hand, may require technical knowledge in programming and in circumstances where a digital prototype is better suited,

limited knowledge in game programming can be a barrier. This lab aims to ease such barriers by providing ingredients and recipes for implementing simple game play mechanics for creating digital game prototypes in Unreal and Unity.

1.1 Audience

The intended audience is aspiring or current game designers with limited or no knowledge of programming who may be interested in being able to test their ideas through digital game prototypes. It is recommended to have some introductory game engine knowledge in Unreal or Unity such as how to create game projects and an understanding of the user interface. Students, game development beginners, or perhaps artists wanting to show their work in a game play context would benefit from the concepts covered in this lab. Not only would it provide tools and concepts for creating game mechanics for digital prototypes, but it would help foster a deeper understanding of working in game engines and maybe even learn a little coding along the way.

2 OUTLINE

The lab is organized into four main parts: (1) introduction, (2) ingredients, (3) recipes, (4) prototyping.

The introduction briefly discusses the importance of prototyping with examples of different types. It also provides examples of prototypes from well-known games, as well as thoughts on prototyping from other game developers.

The ingredients section outlines four main ingredients by defining them generally, then showing examples in both Unreal and Unity. Participants are given access to project files with implementations of these ingredients (Figure 1) which they can follow along with or create their own. The four ingredients are: (1) vectors, (2) time, (3) collisions, (4) lines and rays.

The recipes section shows examples of how to combine the ingredients in ways to create simple mechanics. A few examples are demonstrated live in both Unreal and Unity which participants can follow along if they choose.

The lab concludes with the prototyping section which encourages participants to use the concepts covered in the lab to implement game play mechanics. Participants may choose to work in groups or individually and are given random prototype prompts which they will attempt to create game play mechanics for.

3 CONCLUSION

Prototyping games is an essential part of game design that allows designers to design, develop, and iterate on ideas and concepts of gameplay. Sometimes ideas that seem viable on paper do not translate to game play in a digital format. This lab enables people who may not yet have the programming skills to develop sophisticated gameplay mechanics, to prototype ideas digitally in game engines.

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SIGGRAPH Labs '24, July 27–August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0518-2/24/07

<https://doi.org/10.1145/3641236.3665165>

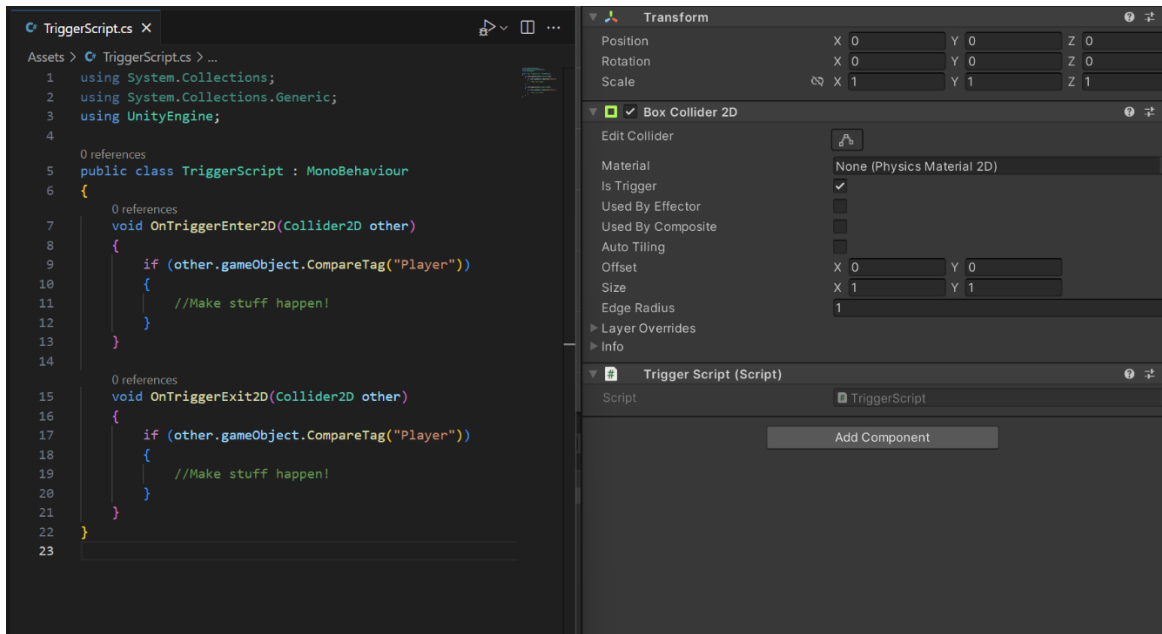


Figure 1: Screenshot from Unity. Image by Author.

Upon completing this lab, participants will have a foundational understanding of developing simple gameplay mechanics in game engines. This could be a starting point to a journey of designing games that allows people to bring their ideas to life, participate in game jams, or even use prototyping for pitches and recruit specialized team members for full development. The flexibility of how people participate is also useful as they can focus on what is useful to them. Last, people will have access to the full notes as well as the Unreal and Unity project files to review for further learning or begin modifying the recipes right away and start designing prototypes.

4 ADDITIONAL RESOURCES

The following are resources related to this lab and are recommended for further learning.

- Unreal Engine's Learning Library by Epic Games <https://dev.epicgames.com/community/unreal-engine/learning>.
- Unity Learning Pathways by Unity <https://learn.unity.com/>.
- Weaving 13 Prototypes into 1 Game: Lessons from Edith Finch by Ian Dallas at GDC 2018 [https://youtu.be/0xVYVP0hxME?si=\\$nxIc4F_9KKtPNkbD](https://youtu.be/0xVYVP0hxME?si=$nxIc4F_9KKtPNkbD).
- Game Design Workshop by Tracy Fullerton